

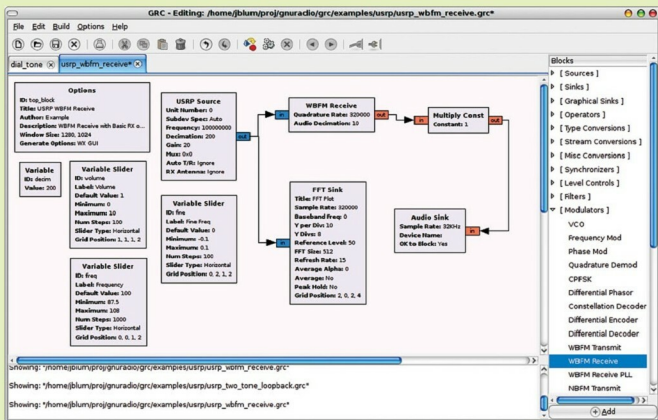


Fig. 2: GNU Radio Companion

Blocks, items and flow graphs

The signal processing takes place inside blocks. Every block has a unique function to perform, which makes the design modular and flexible. Output of the blocks is referred to as an item; an item could be anything with a digital representation.

As in graph theory, the graph through which the data flows is called a flow graph. Blocks are present at the nodes of such graphs and the data flows along the edges of this graph.

performance, frame based processing is often preferred.

In GNU Radio, flow graphs only have item based processing. An item is mostly a sample but could be

a vector too. A logical description of inputs at the input port gives you the item size. In this environment, processing of the maximum possible items takes place every

time the performance of the design is improved.

The downside of GNU Radio approach is that it does not allow introduction of recursive flow graphs.

Creating blocks and extending applications

From being just a beginner to a researcher or professional developing something peculiar, you may feel the lack of certain components. There are also chances that blocks are not

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